

① 베이즈로 사후 $f_{\Theta|X}$ 먼저 ② MMSE=조건부평균 / MAP=argmax 사후 / ML=argmax likelihood ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- LMS=MMSE 점추정: $x^*_M(y) = E[X | Y=y]$ (사후평균 = 최소 MSE 추정량)
- blind/사전추정: 데이터 없으면 $x^*_B = E[X]$ (사전평균)
- 사건조건 MMSE: 구간 사건 A(예: $X > 1/2$)만 알면 $x^* = E[X | A]$
- 최소 MSE: $e^*(y) = \text{Var}(X | Y=y) = E[(X - x^*_M(y))^2 | Y=y]$
- 조건부PDF 먼저: $f_{X|Y}(x|y) = f_{X,Y}(x,y) / f_Y(y)$, $f_Y(y) = \int f_{X,Y}(x,y) dx$
- MAP 점추정: $\theta^*_{\text{MAP}} = \arg\max_{\theta} f_{\Theta|X}(\theta|x) \propto f_{X|\Theta}(x|\theta) \cdot f_{\Theta}(\theta)$ (사후 최대)
- ML 점추정: $\theta^*_{\text{ML}} = \arg\max_{\theta} f_{X|\Theta}(x|\theta)$ (사전 무시); Θ 가 균등(uniform)이면 MAP=ML
- 모든 PMF/PDF는 명시 구간 밖에서 otherwise = 0

문항 목차 (총 5문항)

- [1] L11 · PSP 12.1.3 — [■] 결합PDF·주변 f_X/f_Y ·blind추정 x^*_B ·사건조건 MMSE($X > 1/2$)
- [2] L11 · PSP 12.1.4 — [●] 조건부PDF·MMSE $x^*_M(y) = E[X|Y=y]$ 양방향 핵심정의 ★
- [3] L11 · PSP 12.1.5 — [■] 조건부PDF·MMSE·최소MSE $e^*(0.5) = \text{Var}(X|Y=0.5)$
- [4] L11 · PSP 12.1.6 — [■] 가우시안 신호+잡음·MMSE 추정량 $(Y) \cdot \text{MSE}$ ★
- [5] L11 · PSP 12.3.1 — [■] MAP 추정·MAP vs ML 동일여부 비교 ★

12.1.3 ■ X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2 & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_X(x)$?
- (b) What is the blind estimate $\hat{x}_B$?
- (c) What is the minimum mean square error estimate of X given $X > 1/2$?
- (d) What is $f_Y(y)$?
- (e) What is the blind estimate $\hat{y}_B$?
- (f) What is the minimum mean square error estimate of Y given $X > 1/2$?

12.1.4● X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 6(y-x) & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_{X|Y}(x|y)$?
- (b) What is $\hat{x}_M(y)$, the MMSE estimate of X given $Y = y$?
- (c) What is $f_{Y|X}(y|x)$?
- (d) What is $\hat{y}_M(x)$, the MMSE estimate of Y given $X = x$?

12.1.5 ■ X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2 & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_{X|Y}(x|y)$?
- (b) What is $\hat{x}_M(y)$, the MMSE estimate of X given $Y = y$?
- (c) What is

$$e^*(0.5) = \text{E} \left[(X - \hat{x}_M(0.5))^2 | Y = 0.5 \right],$$

the minimum mean square error of the estimate of X given $Y = 0.5$?

12.1.6■ A signal X and noise Z are independent Gaussian $(0, 1)$ random variables, and $Y = X + Z$ is a noisy observation of the signal X . Usually, we want to use Y to estimate of X ; however, in this problem we will use Y to estimate the noise Z .

- (a) Find $\hat{Z}(Y)$, the MMSE estimator of Z given Y .
- (b) Find the mean squared error $e = E[(Z - \hat{Z}(Y))^2]$.

12.3.1 ■ Suppose that in Quiz 12.3, R , measured in meters, has a uniform PDF over $[0, 1000]$. Find the MAP estimate of R given $X = x$. In this case, are the MAP and ML estimators the same?